

MEHAK VERMA

+91 6261025100 missmehak755@gmail.com github.com/Mehak-991 mehak-verma99

Education

B.Tech in Information Technology

Sept 2023 – Aug 2027

Experience

TechieShubhdeep IT Solutions Pvt Ltd — Data Science intern - Certificate

Present 2026

On-site, Gwalior

- Engineered and fine-tuned ML models using Python and Scikit-learn, achieving 15–20% increase in model accuracy through feature engineering and predictive analysis.
- Performed data preprocessing and exploratory data analysis on 500+ records, generating 25% more actionable insights via statistical analysis to enhance model performance.
- Generated insightful analytical reports and visualizations with Pandas, NumPy, Matplotlib, and Seaborn, reducing manual data processing time by more than 8+ hours each week.

Cognifyz Technologies — Machine Learning Intern - Certificate

Jun 2025 – Jul 2025

Remote

- Performed end-to-end data analysis on 10,000+ records using Python (Pandas, NumPy) to generate actionable business insights and support data-driven decision-making.
- Applied statistical analysis, data cleaning, and feature interpretation techniques to solve 15+ real-world business problems using structured and unstructured datasets.
- Built analytical reports and dashboards using data visualization tools (Matplotlib, Seaborn) to improve insight accuracy and decision efficiency by 20%.

Projects

Movie-Recommendation-System-using-NLP-TF-IDF — Python, Scikit-Learn, Django, ML

May 2026

github.com/Movie-Recommendation-System-using-NLP-TF-IDF.git

- Constructed a robust Machine learning workflow utilizing Pandas, Scikit-learn, and Parquet for processing over 1,000 records, achieving 40% processing efficiency gains.
- Developed a semantic recommendation engine using NLP and TF-IDF vectorization to extract meaningful patterns from 10,000+ unstructured movie metadata records.
- Optimized inference performance using Truncated SVD and sparse matrix (.npz) representations, reducing memory usage by 50% and achieving sub-50ms query latency.

California House Price Prediction Model — Python, scikit-learn, Pandas, NumPy, Gradio

Jan 2026

github.com/Mehak-991/California-House-Price-Prediction-Model.git

- Built a Random Forest regression model to predict real estate prices using geographical and financial features on 20,000+ records with high accuracy.
- Engineered automated preprocessing pipelines using ColumnTransformer for imputation, scaling, and one-hot encoding on 20,000+ records.
- Analyzed a real-time interactive ML web application using Gradio, enabling instant property price inference with low-latency model serving and user-friendly UI.

SMS-Spam-Classification-System-using-NLP — ML, NLTK, Streamlit, Pickle, Naive Bayes

Nov 2025

github.com/Mehak-991/SMS-Spam-Classification-System-using-NLP.git

- Developed an NLP-based text classification pipeline using NLTK for tokenization, stemming, and text preprocessing on 5,000+ messages, improving text processing efficiency and model readiness for machine learning tasks.
- Formulated a precise spam detection model using TF-IDF and Naive Bayes, achieving 97.10% accuracy in identifying unwanted messages.
- Revamped a production Streamlit web application, enabling real-time user text classification and cutting user interaction steps by 20% through UI improvements.

Skills

Languages: Python, C++, JavaScript

Frontend: React.js, HTML5, CSS3.

ML: Transformers, Hugging Face, Vision Transformer, TensorFlow, PyTorch

Analytics: EDA, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Statistical Analysis

Frameworks: Django, Streamlit, Gradio, Keras

Tools: Git, GitHub, Docker, Google Colab, Postman, Jupyter Notebook

Databases / Cloud: PostgreSQL, MySQL, MongoDB, Pinecone, AWS, Render, Vercel, Streamlit Cloud

Generative & Agentic AI: LangChain, LangGraph, CrewAI, RAG

Achievement

- 1× Hackathon Certificate - Participated in Web3 Hackathon and TantraFiesta'25 organized by IIT Nagpur - Certificates
- Authored research paper titled *“Machine Learning-driven Heart Disease Detection with Flask Based Application Deployment”*; currently under academic review for publication.